

ENGINEERING MATHEMATICS • B.TECH 1ST YEAR • 1ST SEMESTER

# VECTOR DIFFERENTIATION

## UNIT IV

*Complete Formula Sheet • Last-Day Revision • Exam-Ready*

### SUBJECT

Engineering Mathematics

### UNIT

Unit IV

### TOPICS

$\nabla$ , Grad, Div, Curl

### PAGES

10–15 Pages

01

Course Objectives

02

Basic Definitions

03

Del Operator  $\nabla$

04

Gradient  $\nabla\phi$

05

Directional Derivative

06

Divergence  $\nabla\cdot\mathbf{F}$

07

Curl  $\nabla\times\mathbf{F}$

08

Vector Identities

09

Comparison Tables

10

Flashcards (30 Q&A)

11

Exam Tips

12

Quick Revision Sheet

$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$  • Your Formula Weapon for Today's Exam

## 01 SECTION ONE COURSE OBJECTIVES

- ✓ Understand scalar and vector point functions and their domains
- ✓ Compute gradient, divergence, and curl of given functions
- ✓ Use vector identities for algebraic simplification
- ✓ Apply the vector differential operator ( $\text{Del} / \nabla$ ) correctly
- ✓ Interpret the physical meaning of each vector operation
- ✓ Identify solenoidal and irrotational vector fields

## 02 SECTION TWO BASIC DEFINITIONS

### SCALAR POINT FUNCTION

A function  $\psi(x, y, z)$  that assigns a **single real number** (magnitude only) to each point in space.

*Examples: Temperature  $T(x,y,z)$ , pressure  $P(x,y,z)$*

$$\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$$

### VECTOR POINT FUNCTION

A function  $F(x, y, z)$  that assigns a **vector** to each point in space.

*Examples: Velocity  $V(x,y,z)$ , force  $F(x,y,z)$*

$$F = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$$

### SCALAR FIELD

A region where a scalar point function is defined at every point.

### VECTOR FIELD

A region where a vector point function is defined at every point.

### POSITION VECTOR

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

$$|\mathbf{r}| = r = \sqrt{x^2 + y^2 + z^2}$$

### UNIT VECTORS

| Symbol       | Direction    | Magnitude |
|--------------|--------------|-----------|
| $\mathbf{i}$ | Along x-axis | 1         |
| $\mathbf{j}$ | Along y-axis | 1         |
| $\mathbf{k}$ | Along z-axis | 1         |

**KEY DISTINCTION** **Scalar Field** — only magnitude | **Vector Field** — magnitude + direction

Independent variables:  $x, y, z$  (spatial coordinates in 3D space)

## 03 SECTION THREE VECTOR DIFFERENTIAL OPERATOR – DEL ( $\nabla$ )

### DEFINITION OF DEL OPERATOR

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

- $\nabla$  is a **vector differential operator**, not a vector by itself
- It **operates on** scalar or vector functions
- Read as: "*del*" or "*nabla*"

| Operation                  | Symbol                    | Applied To                   | Result Type | Name       |
|----------------------------|---------------------------|------------------------------|-------------|------------|
| $\nabla\phi$               | $\text{grad } \phi$       | Scalar function $\phi$       | Vector      | Gradient   |
| $\nabla \cdot \mathbf{F}$  | $\text{div } \mathbf{F}$  | Vector function $\mathbf{F}$ | Scalar      | Divergence |
| $\nabla \times \mathbf{F}$ | $\text{curl } \mathbf{F}$ | Vector function $\mathbf{F}$ | Vector      | Curl       |
| $\nabla^2\phi$             | Laplacian $\phi$          | Scalar function $\phi$       | Scalar      | Laplacian  |

### Laplacian Operator

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} \text{ (scalar result)}$$

## 04 SECTION FOUR GRADIENT ( $\nabla\phi$ )

### DEFINITION

The **gradient of a scalar function  $\phi$**  is a vector that points in the direction of maximum rate of increase of  $\phi$ .

$$\nabla\phi = \text{grad } \phi = \mathbf{i} \frac{\partial\phi}{\partial x} + \mathbf{j} \frac{\partial\phi}{\partial y} + \mathbf{k} \frac{\partial\phi}{\partial z}$$

### GEOMETRIC MEANING

- $\nabla\phi$  is **normal** (perpendicular) to the surface  $\phi = \text{constant}$
- Points in direction of **steepest ascent**
- $|\nabla\phi|$  = magnitude of maximum rate of increase

### KEY PROPERTIES OF GRADIENT

| Property              | Result                                     |
|-----------------------|--------------------------------------------|
| $\nabla(\phi + \psi)$ | $\nabla\phi + \nabla\psi$                  |
| $\nabla(\phi\psi)$    | $\phi\nabla\psi + \psi\nabla\phi$          |
| $\nabla(\phi/\psi)$   | $(\psi\nabla\phi - \phi\nabla\psi)/\psi^2$ |
| $\nabla(r^n)$         | $n \cdot r^{n-2} \cdot \mathbf{r}$         |
| $\nabla(1/r)$         | $-\mathbf{r}/r^3 = -\hat{\mathbf{r}}/r^2$  |

### NORMAL TO A SURFACE

Unit normal to surface  $\phi(x, y, z) = c$  :

$$\hat{\mathbf{n}} = \nabla\phi / |\nabla\phi|$$

## 05 SECTION FIVE DIRECTIONAL DERIVATIVE

### DEFINITION

The rate of change of  $\phi$  in the direction of a given unit vector  $\hat{a}$ .

$$D_{\hat{a}} \phi = \nabla \phi \cdot \hat{a}$$

where  $\hat{a}$  = **unit vector** in desired direction.

### HOW TO FIND UNIT VECTOR $\hat{a}$

If direction is given by vector  $a = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ :

$$\hat{a} = a / |a| = a / \sqrt{(a_1^2 + a_2^2 + a_3^2)}$$

### IMPORTANT RESULTS

- **Maximum value:**  $|\nabla \phi|$  — in the gradient direction
- **Minimum value:**  $-|\nabla \phi|$  — opposite to gradient
- **Zero:** when  $\hat{a} \perp \nabla \phi$  (perpendicular direction)

### ⚠ COMMON EXAM MISTAKE

Always divide direction vector by its magnitude to get **unit vector  $\hat{a}$**  before applying the formula. Forgetting this is the #1 error!

| Direction                      | Directional Derivative                             | Value                             |
|--------------------------------|----------------------------------------------------|-----------------------------------|
| Along $\nabla \phi$            | $\nabla \phi \cdot (\nabla \phi /  \nabla \phi )$  | <b>Maximum</b> = $ \nabla \phi $  |
| Opposite to $\nabla \phi$      | $\nabla \phi \cdot (-\nabla \phi /  \nabla \phi )$ | <b>Minimum</b> = $- \nabla \phi $ |
| Perpendicular to $\nabla \phi$ | $\nabla \phi \cdot \hat{a} = 0$                    | <b>Zero</b>                       |

## 06 SECTION SIX DIVERGENCE ( $\nabla \cdot F$ )

### FORMULA

If  $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ :

$$\nabla \cdot \mathbf{F} = \operatorname{div} \mathbf{F} = \partial F_1 / \partial x + \partial F_2 / \partial y + \partial F_3 / \partial z$$

- Input: **Vector function  $\mathbf{F}$**
- Output: **Scalar**

### SOLENOIDAL VECTOR FIELD

A vector field is **solenoidal** (no source/sink) if:

$$\nabla \cdot \mathbf{F} = 0$$

*Example: Magnetic field  $\mathbf{B}$  is solenoidal*

### PHYSICAL MEANING OF DIVERGENCE

- $\nabla \cdot \mathbf{F} > 0 \rightarrow$  Source (fluid flowing out)
- $\nabla \cdot \mathbf{F} < 0 \rightarrow$  Sink (fluid flowing in)
- $\nabla \cdot \mathbf{F} = 0 \rightarrow$  Incompressible / solenoidal

Think: "divergence measures how much a vector field spreads out from a point"

### DIVERGENCE PROPERTIES

- $\nabla \cdot (\mathbf{F} + \mathbf{G}) = \nabla \cdot \mathbf{F} + \nabla \cdot \mathbf{G}$
- $\nabla \cdot (\psi \mathbf{F}) = \psi \nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla \psi$
- $\nabla \cdot (\nabla \psi) = \nabla^2 \psi$  (Laplacian)
- $\nabla \cdot (\nabla \times \mathbf{F}) = 0$  (always)

## 07 SECTION SEVEN CURL ( $\nabla \times \mathbf{F}$ )

### DEFINITION

Measures the **rotational tendency** (circulation) of vector field  $\mathbf{F}$  at a point.

- Input: **Vector function  $\mathbf{F}$**
- Output: **Vector**

### IRROTATIONAL VECTOR FIELD

A vector field is **irrotational** (no rotation) if:

$$\nabla \times \mathbf{F} = \mathbf{0}$$

An irrotational field is also called a **conservative field** and can be expressed as a gradient:  $\mathbf{F} = \nabla \phi$

### PHYSICAL MEANING

- $\nabla \times \mathbf{F} \neq \mathbf{0} \rightarrow$  Rotating / circulating field
- $\nabla \times \mathbf{F} = \mathbf{0} \rightarrow$  Irrotational / conservative
- Direction of  $\nabla \times \mathbf{F}$  = axis of rotation (right-hand rule)
- $|\nabla \times \mathbf{F}|$  = angular velocity  $\times 2$

### DETERMINANT FORMULA

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_1 & F_2 & F_3 \end{vmatrix}$$

### EXPANDED FORM

$$\nabla \times \mathbf{F} = \mathbf{i}(\partial F_3/\partial y - \partial F_2/\partial z) - \mathbf{j}(\partial F_3/\partial x - \partial F_1/\partial z) + \mathbf{k}(\partial F_2/\partial x - \partial F_1/\partial y)$$

### ⚠ SIGN PATTERN — REMEMBER!

| Component    | Formula                                             | Sign of $\mathbf{j}$ |
|--------------|-----------------------------------------------------|----------------------|
| $\mathbf{i}$ | $\partial F_3/\partial y - \partial F_2/\partial z$ | $+i$                 |
| $\mathbf{j}$ | $\partial F_3/\partial x - \partial F_1/\partial z$ | $-j$                 |
| $\mathbf{k}$ | $\partial F_2/\partial x - \partial F_1/\partial y$ | $+k$                 |

The middle term ( $\mathbf{j}$  component) is **NEGATIVE** — most common mistake!

### Curl Properties

- $\nabla \times (\mathbf{F} + \mathbf{G}) = \nabla \times \mathbf{F} + \nabla \times \mathbf{G}$
- $\nabla \times (\nabla \phi) = \mathbf{0}$  (curl of gradient = 0)
- $\nabla \times (\phi \mathbf{F}) = \phi (\nabla \times \mathbf{F}) + (\nabla \phi) \times \mathbf{F}$
- $\nabla \cdot (\nabla \times \mathbf{F}) = \mathbf{0}$  (div of curl = 0)

## GRADIENT IDENTITIES

| # | Identity                                                                                                              |
|---|-----------------------------------------------------------------------------------------------------------------------|
| 1 | $\nabla(\varphi + \psi) = \nabla\varphi + \nabla\psi$                                                                 |
| 2 | $\nabla(\varphi\psi) = \varphi\nabla\psi + \psi\nabla\varphi$                                                         |
| 3 | $\nabla(\varphi/\psi) = (\psi\nabla\varphi - \varphi\nabla\psi) / \psi^2$                                             |
| 4 | $\nabla(F \cdot G) = (F \cdot \nabla)G + (G \cdot \nabla)F + F \times (\nabla \times G) + G \times (\nabla \times F)$ |

## DIVERGENCE IDENTITIES

| # | Identity                                                                            |
|---|-------------------------------------------------------------------------------------|
| 1 | $\nabla \cdot (F+G) = \nabla \cdot F + \nabla \cdot G$                              |
| 2 | $\nabla \cdot (\varphi F) = \varphi(\nabla \cdot F) + F \cdot (\nabla \varphi)$     |
| 3 | $\nabla \cdot (F \times G) = G \cdot (\nabla \times F) - F \cdot (\nabla \times G)$ |
| 4 | $\nabla \cdot (\nabla \varphi) = \nabla^2 \varphi$ (Laplacian)                      |
| 5 | $\nabla \cdot (\nabla \times F) = 0$ (always zero)                                  |

## CURL IDENTITIES

| # | Identity                                                                                                     |
|---|--------------------------------------------------------------------------------------------------------------|
| 1 | $\nabla \times (F+G) = \nabla \times F + \nabla \times G$                                                    |
| 2 | $\nabla \times (\varphi F) = \varphi(\nabla \times F) + (\nabla \varphi) \times F$                           |
| 3 | $\nabla \times (F \times G) = F(\nabla \cdot G) - G(\nabla \cdot F) + (G \cdot \nabla)F - (F \cdot \nabla)G$ |
| 4 | $\nabla \times (\nabla \varphi) = 0$ (curl of gradient = 0)                                                  |
| 5 | $\nabla \times (\nabla \times F) = \nabla(\nabla \cdot F) - \nabla^2 F$                                      |

## MOST CRITICAL STANDARD RESULTS

| Result                           | Value              | Meaning                  |
|----------------------------------|--------------------|--------------------------|
| $\nabla \cdot (\nabla \varphi)$  | $\nabla^2 \varphi$ | Laplacian                |
| $\nabla \times (\nabla \varphi)$ | 0                  | Curl of gradient = 0     |
| $\nabla \cdot (\nabla \times F)$ | 0                  | Div of curl = 0          |
| $\nabla r$                       | $\hat{r} = r/r$    | Unit position vector     |
| $\nabla(1/r)$                    | $-r/r^3$           | Used in potential theory |
| $\nabla \cdot r$                 | 3                  | Divergence of r          |
| $\nabla \times r$                | 0                  | Curl of r = 0            |

09

SECTION NINE

COMPARISON TABLES

GRADIENT VS DIVERGENCE VS CURL

| Feature          | Gradient ( $\nabla\phi$ )                                                        | Divergence ( $\nabla\cdot F$ )                                                | Curl ( $\nabla\times F$ )          |
|------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------|
| Input            | Scalar $\phi$                                                                    | Vector $F$                                                                    | Vector $F$                         |
| Output           | Vector                                                                           | Scalar                                                                        | Vector                             |
| Symbol           | $\nabla\phi$                                                                     | $\nabla\cdot F$                                                               | $\nabla\times F$                   |
| Physical meaning | Max rate of increase                                                             | Source/sink strength                                                          | Rotation/circulation               |
| Zero condition   | $\phi = \text{constant}$                                                         | Solenoidal: $\nabla\cdot F = 0$                                               | Irrotational: $\nabla\times F = 0$ |
| Operation type   | Del $\cdot$ scalar                                                               | Dot product                                                                   | Cross product                      |
| Formula          | $i\partial\phi/\partial x + j\partial\phi/\partial y + k\partial\phi/\partial z$ | $\partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$ | Determinant form                   |

SCALAR FIELD VS VECTOR FIELD

| Property       | Scalar Field            | Vector Field                    |
|----------------|-------------------------|---------------------------------|
| Represented by | $\phi(x,y,z)$           | $F(x,y,z)$                      |
| Output         | Single number           | Vector (mag+dir)                |
| Example        | Temp, Pressure          | Velocity, Force                 |
| Operations     | $\nabla\phi$ (gradient) | $\nabla\cdot F, \nabla\times F$ |
| Visualized as  | Level surfaces          | Field lines                     |

SOLENOIDAL VS IRRATIONAL

| Property         | Solenoidal          | Irrotational         |
|------------------|---------------------|----------------------|
| Condition        | $\nabla\cdot F = 0$ | $\nabla\times F = 0$ |
| Operation        | Divergence = 0      | Curl = 0             |
| Physics          | No source/sink      | No rotation          |
| Also called      | Source-free         | Conservative         |
| Example          | Magnetic field $B$  | Electrostatic $E$    |
| Scalar potential | —                   | $F = \nabla\phi$     |

QUICK OPERATOR SUMMARY

| Operator   | Notation         | Reads As        | Input $\rightarrow$ Output  | Key Formula                                                                               |
|------------|------------------|-----------------|-----------------------------|-------------------------------------------------------------------------------------------|
| Del        | $\nabla$         | nabla / del     | Operator                    | $i\partial/\partial x + j\partial/\partial y + k\partial/\partial z$                      |
| Gradient   | $\nabla\phi$     | grad phi        | Scalar $\rightarrow$ Vector | $i\partial\phi/\partial x + j\partial\phi/\partial y + k\partial\phi/\partial z$          |
| Divergence | $\nabla\cdot F$  | div F           | Vector $\rightarrow$ Scalar | $\partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$             |
| Curl       | $\nabla\times F$ | curl F          | Vector $\rightarrow$ Vector | 3 $\times$ 3 determinant                                                                  |
| Laplacian  | $\nabla^2\phi$   | del-squared phi | Scalar $\rightarrow$ Scalar | $\partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$ |

# 10 SECTION TEN — EXAM Q&A

## FLASHCARDS — 30 Q&A

### Q01 What is a scalar field?

A function that assigns a **single real number** (magnitude only) to each point in space. E.g.,  $\phi(x,y,z)$  — temperature distribution.

### Q03 Define the Del operator.

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

A vector differential operator — not a vector itself.

### Q05 What does gradient represent geometrically?

$\nabla\phi$  is the **normal vector** to the surface  $\phi = \text{constant}$ . It points in direction of steepest ascent.

### Q07 Maximum value of directional derivative?

$|\nabla\phi|$  — occurs in the direction of the gradient itself.

### Q09 Divergence formula?

$$\nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Input: vector. Output: **scalar**.

### Q11 Condition for solenoidal field?

$\nabla \cdot \mathbf{F} = 0$  — no source or sink. Also called source-free or divergence-free field.

### Q13 What does curl represent?

Measures the **rotational tendency** (circulation) of the vector field at a point. Direction = axis of rotation.

### Q15 What is the Laplacian operator?

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\nabla^2\phi = \nabla \cdot (\nabla\phi)$$

### Q17 What is $\nabla \cdot (\nabla \times \mathbf{F})$ ?

$\nabla \cdot (\nabla \times \mathbf{F}) = 0$  — Divergence of any curl is always **zero**.

### Q19 What is $\nabla \times \mathbf{r}$ ?

$\nabla \times \mathbf{r} = 0$  — position vector  $\mathbf{r}$  is always irrotational.

### Q21 Identity: $\nabla \cdot (\phi \mathbf{F}) = ?$

$$\nabla \cdot (\phi \mathbf{F}) = \phi (\nabla \cdot \mathbf{F}) + \mathbf{F} \cdot (\nabla\phi)$$

Product rule for divergence.

### Q23 Unit normal to surface $\phi = c$ ?

### Q02 What is a vector field?

A function that assigns a **vector** (magnitude + direction) to each point in space. E.g.,  $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$  — velocity field.

### Q04 What is the gradient of $\phi$ ?

$$\nabla\phi = \mathbf{i} \frac{\partial\phi}{\partial x} + \mathbf{j} \frac{\partial\phi}{\partial y} + \mathbf{k} \frac{\partial\phi}{\partial z}$$

Points in direction of maximum increase of  $\phi$ .

### Q06 Directional derivative formula?

$$D_{\hat{\mathbf{a}}} \phi = \nabla\phi \cdot \hat{\mathbf{a}}$$

where  $\hat{\mathbf{a}}$  is the **unit** vector in the desired direction.

### Q08 When is directional derivative zero?

When the direction vector  $\hat{\mathbf{a}}$  is **perpendicular to  $\nabla\phi$**  (i.e., along the level surface  $\phi = \text{constant}$ ).

### Q10 Physical meaning of divergence?

Measures **source/sink** strength.  $\nabla \cdot \mathbf{F} > 0$  = source;  $\nabla \cdot \mathbf{F} < 0$  = sink;  $\nabla \cdot \mathbf{F} = 0$  = incompressible.

### Q12 Curl formula (determinant)?

$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$  — expand as 3×3 determinant.

### Q14 Condition for irrotational field?

$\nabla \times \mathbf{F} = 0$  — no rotation. Also called conservative field. Then  $\mathbf{F} = \nabla\phi$  for some scalar  $\phi$ .

### Q16 What is $\nabla \times (\nabla\phi)$ ?

$\nabla \times (\nabla\phi) = 0$  — Curl of any gradient is always **zero**.

### Q18 What is $\nabla \cdot \mathbf{r}$ ?

$\nabla \cdot \mathbf{r} = 3$ , where  $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$  is the position vector.

### Q20 What is $\nabla(1/r)$ ?

$$\nabla(1/r) = -\mathbf{r}/r^3 = -\hat{\mathbf{r}}/r^2$$

### Q22 Identity: $\nabla \times (\phi \mathbf{F}) = ?$

$$\nabla \times (\phi \mathbf{F}) = \phi (\nabla \times \mathbf{F}) + (\nabla\phi) \times \mathbf{F}$$

Product rule for curl.

### Q24 Gradient acts on \_\_\_\_, produces \_\_\_\_?

$\hat{n} = \nabla\psi / |\nabla\psi|$  — gradient direction normalized to unit length.

Gradient acts on a **scalar** function and produces a **vector** function. Scalar in  $\rightarrow$  Vector out.

**Q25** Divergence acts on \_\_\_\_, produces \_\_\_\_?

Divergence acts on a **vector** function and produces a **scalar**. Vector in  $\rightarrow$  Scalar out.

**Q26** Curl acts on \_\_\_\_, produces \_\_\_\_?

Curl acts on a **vector** function and produces a **vector**. Vector in  $\rightarrow$  Vector out.

**Q27** What is a conservative field?

A vector field  $F$  where  $\nabla \times F = 0$ . Can be written as  $F = \nabla\phi$  for some scalar potential  $\phi$ .

**Q28**  $\nabla(r^n) = ?$

$\nabla(r^n) = n \cdot r^{n-2} \cdot r$  — used in potential theory and radial functions.

**Q29**  $\nabla^2(1/r) = ?$

$\nabla^2(1/r) = 0$  for  $r \neq 0$  —  $1/r$  is a harmonic function (satisfies Laplace's equation).

**Q30** Minimum directional derivative value?

$-|\nabla\psi|$  — in the direction **opposite** to the gradient (steepest descent direction).

### ⚠ CRITICAL MISTAKES TO AVOID

- ✗ **Forgetting unit vector in directional derivative.**  
The formula is  $\nabla\phi \cdot \hat{a}$ , not  $\nabla\phi \cdot a$ . Always normalize the direction vector first.

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- ✗ **Sign error in curl (j component).** The j component has a minus sign:  $\nabla \times F = i(\dots) - j(\dots) + k(\dots)$ . Middle term is negative!

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- ✗ **Confusing gradient with directional derivative.**  
Gradient = vector; Directional derivative = scalar (dot product with unit vector).

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- ✗ **Writing  $\nabla \cdot F$  as a vector.** Divergence is a scalar. If your answer is a vector, you've computed curl or gradient instead.

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- ✗ **Writing  $\nabla\phi$  as a scalar.** Gradient is a vector. Your answer must have i, j, k components.

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- ✗ **Forgetting Laplacian formula.**  $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$  — second-order partial derivatives, not first.

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- ✗ **Mixing solenoidal and irrotational conditions.**  
Solenoidal  $\rightarrow \nabla \cdot F = 0$  (divergence). Irrotational  $\rightarrow \nabla \times F = 0$  (curl). Don't swap them!

### ✓ EXAM STRATEGY — SCORE MORE

- ★ **For gradient problems:** Differentiate each component separately. Check all three:  $\partial\phi/\partial x$ ,  $\partial\phi/\partial y$ ,  $\partial\phi/\partial z$ .

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- ★ **For curl problems:** Write out the  $3 \times 3$  determinant first, then expand row by row. Don't skip the middle minus sign.

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- ★ **For directional derivative:** Step 1: find  $\nabla\phi$ . Step 2: find  $|\text{direction vector}|$ . Step 3: compute  $\hat{a} = a/|a|$ . Step 4: dot product.

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- ★ **For proving solenoidal/irrotational:** Compute  $\nabla \cdot F$  or  $\nabla \times F$  and show it equals 0. State the conclusion clearly.

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- ★ **Using vector identities:** Memorize  $\nabla \cdot (\nabla \times F) = 0$  and  $\nabla \times (\nabla\phi) = 0$ . These save time in proof questions.

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- ★ **Normal to surface:** If asked for unit normal — always divide  $\nabla\phi$  by  $|\nabla\phi|$ .

### 💡 MEMORY TRICKS

- ▶ **GVS:** Gradient  $\rightarrow$  Vector output | DiVerge  $\rightarrow$  Scalar | Curl  $\rightarrow$  Vector
- ▶ **SD  $\leftrightarrow$  IC:** Solenoidal = Divergence=0 | Irrotational = Curl=0
- ▶ **Curl minus j:** "i plus, j MINUS, k plus" — never forget the middle sign
- ▶  **$\nabla^2$ :** Laplace = second derivatives, not first!
- ▶ **Unit vector FIRST:** Before dot product in directional derivative

# ⚡ QUICK REVISION SHEET — ALL FORMULAS AT A GLANCE ⚡

## DEL OPERATOR

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

## GRADIENT

$$\nabla \psi = \mathbf{i} \frac{\partial \psi}{\partial x} + \mathbf{j} \frac{\partial \psi}{\partial y} + \mathbf{k} \frac{\partial \psi}{\partial z}$$

Scalar  $\rightarrow$  Vector  $\cdot$  Direction of max increase

## DIRECTIONAL DERIVATIVE

$$D_{\hat{a}} \psi = \nabla \psi \cdot \hat{a}$$

Max =  $|\nabla \psi|$   $\cdot$  Min =  $-|\nabla \psi|$   $\cdot$  Zero when  $\perp$  to  $\nabla \psi$

## UNIT NORMAL

$$\hat{n} = \nabla \psi / |\nabla \psi|$$

## DIVERGENCE

$$\nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Vector  $\rightarrow$  Scalar  $\cdot$  Solenoidal:  $\nabla \cdot \mathbf{F} = 0$

## LAPLACIAN

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2}$$

=  $\nabla \cdot (\nabla \psi)$   $\cdot$  Scalar  $\rightarrow$  Scalar

## CURL

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

Vector  $\rightarrow$  Vector  $\cdot$  Irrotational:  $\nabla \times \mathbf{F} = 0$

## CURL EXPANDED

$$= \mathbf{i}(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}) - \mathbf{j}(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}) + \mathbf{k}(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y})$$

## CRITICAL IDENTITIES

$$\begin{aligned} \nabla \times (\nabla \psi) &= 0 & \nabla \cdot (\nabla \times \mathbf{F}) &= 0 & \nabla \cdot (\nabla \psi) &= \nabla^2 \psi \\ \nabla \cdot (\psi \mathbf{F}) &= \psi \nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla \psi & \nabla \times (\psi \mathbf{F}) &= \psi \nabla \times \mathbf{F} + \nabla \psi \times \mathbf{F} \end{aligned}$$

## STANDARD RESULTS

$$\begin{aligned} \nabla \cdot \mathbf{r} &= 3 & \nabla \times \mathbf{r} &= 0 & \nabla r &= \hat{r} & \nabla(1/r) &= -\mathbf{r}/r^3 \\ \nabla(r^n) &= n \cdot r^{n-2} \cdot \mathbf{r} \end{aligned}$$

## MASTER REFERENCE TABLE

| Concept     | Formula                                                                                                                                                 | I/O               | Zero Condition                | Name when Zero         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------|------------------------|
| Gradient    | $\nabla \psi = \mathbf{i} \frac{\partial \psi}{\partial x} + \mathbf{j} \frac{\partial \psi}{\partial y} + \mathbf{k} \frac{\partial \psi}{\partial z}$ | S $\rightarrow$ V | $\psi = \text{constant}$      | Uniform field          |
| Dir. Deriv. | $\nabla \psi \cdot \hat{a}$                                                                                                                             | S $\rightarrow$ S | $\hat{a} \perp \nabla \psi$   | No change in direction |
| Divergence  | $\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$                                                   | V $\rightarrow$ S | $\nabla \cdot \mathbf{F} = 0$ | Solenoidal             |

| Concept   | Formula                                                                                                                                                       | I/O                                 | Zero Condition                 | Name when Zero    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------|-------------------|
| Curl      | $\det \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \end{vmatrix} \cdot \mathbf{F}$ | $\mathbf{V} \rightarrow \mathbf{V}$ | $\nabla \times \mathbf{F} = 0$ | Irrotational      |
| Laplacian | $\partial^2 \psi / \partial x^2 + \partial^2 \psi / \partial y^2 + \partial^2 \psi / \partial z^2$                                                            | $\mathbf{S} \rightarrow \mathbf{S}$ | $\nabla^2 \phi = 0$            | Harmonic function |

ALL THE BEST FOR YOUR  
EXAM! ✨

$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$